# A Counterexample and Repair for the Annular Lemma in
## Fractional Poincaré Inequalities for General Measures

Autonomous proof packet for `fa_banach_001`

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**Status.** Substantial partial result, likely valid, pending human review. The printed annular estimate is disproved already in the source's $L^2$ setting and the main $L^2$ proof is repaired. The intended global $L^p$ comparison remains open for a precisely identified vector-valued reason.

## 1 Source statement and question

Let $Q(a,s)$ denote the cube centered at $a$ with side length $s$. For a unit-scale covering cube $Q(x_j^t,\sqrt{t})$, the source subtracts the Lebesgue mean of $f$ on $Q(x_j^t, 2\sqrt{t})$:

$$m^{j,t} = \frac{1}{|Q(x_j^t, 2\sqrt{t})|}\int_{Q(x_j^t,2\sqrt{t})} f(y)\,dy, \qquad g^{j,t} = f - m^{j,t}.$$

It then restricts $g^{j,t}$ to

$$C_k^{j,t} = Q(x_j^t, 2^{k+1}\sqrt{t}) \setminus Q(x_j^t, 2^k\sqrt{t}), \qquad g_k^{j,t} = g^{j,t}\mathbf{1}_{C_k^{j,t}}.$$

Assertion B of Lemma 3.4 claims an estimate with the large-cube normalization $(2^k\sqrt{t})^{-n}$.



We claim that:

**Lemma 3.4.** *There exists $\bar{C} > 0$ such that, for all $t > 0$ and all $j \in \mathbb{N}$:*

**A.** *For the first term:*

$$\left\|g_0^{j,t}\right\|^2_{L^2(C_0^{j,t},M)} \le \frac{\bar{C}}{t^{n/2}} \int_{Q\left(x_j^t,2\sqrt{t}\right)}\int_{Q\left(x_j^t,2\sqrt{t}\right)} |f(x)-f(y)|^2\, M(x)\,dx\,dy.$$

**B.** *For all $k \ge 1$,*

$$\left\|g_k^{j,t}\right\|^2_{L^2(C_k^{j,t},M)} \le$$

$$\frac{\bar{C}}{(2^k\sqrt{t})^n}\int_{x\in Q(x_j^t,2^{k+1}\sqrt{t})}\int_{y\in Q(x_j^t,2^{k+1}\sqrt{t})} |f(x)-f(y)|^2\, M(x)\,dx\,dy.$$

We postpone the proof to the end of the section and finish the proof of Lemma 3.3. Using Assertion **A** in Lemma 3.4, summing up on $j \ge 0$

Source PDF p. 11: complete statement of Lemma 3.4, including B.

The printed proof says only that B follows similarly from the calculation for the central cube. The following paragraph asks whether an $L^p$ version of the resulting fractional comparison holds.



which shows Assertion **A**. We argue similarly for Assertion **B** and obtain

$$\left\|g_k^{j,t}\right\|^2_{L^2(C_k^{j,t},M)} \le \frac{C}{2^{k/n}t^{n/2}} \int_{x\in Q(x_j^t,2^{k+1}\sqrt{t})} \int_{y\in Q(x_j^t,2^{k+1}\sqrt{t})} |f(x)-f(y)|^2 \, M(x)\,dx\,dy,$$

which ends the proof of Lemma 3.4. □

We end up this section with a few comments on Lemma 3.4. It is a well-known fact ([Str67]) that, when $0<\alpha<2$, for all $p\in(1,+\infty)$,

$$\left\|(-\Delta)^{\alpha/4}f\right\|_{L^p(\mathbb{R}^n)} \le C_{\alpha,p}\left\|S_\alpha f\right\|_{L^p(\mathbb{R}^n)} \tag{3.22}$$

where

$$S_\alpha f(x) = \left(\int_0^{+\infty}\left(\int_B |f(x+ry)-f(x)|\,dy\right)^2 \frac{dr}{r^{1+\alpha}}\right)^{\frac12},$$

and also ([Ste61])

$$\left\|(-\Delta)^{\alpha/4}f\right\|_{L^p(\mathbb{R}^n)} \le C_{\alpha,p}\left\|D_\alpha f\right\|_{L^p(\mathbb{R}^n)} \tag{3.23}$$

where

$$D_\alpha f(x) = \left(\int_{\mathbb{R}^n} \frac{|f(x+y)-f(x)|^2}{|y|^{n+\alpha}}dy\right)^{\frac12}.$$

In [CRTN01], these inequalities were extended to the setting of a unimodular Lie group endowed with a sub-laplacian $\Delta$, relying on semigroups techniques and Littlewood-Paley-Stein functionals. In particular, in [CRTN01], we use *pointwise* estimates of the kernel of the semigroup generated by $\Delta$. The conclusion of Lemma 3.4 means that the norm of $L^{\alpha/4}f$ in $L^2(\mathbb{R}^n,M)$ is bounded from above by the $L^2(\mathbb{R}^n,M)$ norm of an appropriate version of $D_\alpha$. Note that this does not require pointwise estimates for the kernel of the semigroup generated by $L$, and that the $L^2$ off-diagonal estimates given by Lemma 2.1, which hold for a general measure $M$, are enough for our argument to hold. However, we do not know if an $L^p$ version of Lemma 3.4 still holds. Note also that we do not compare the $L^2(\mathbb{R}^n,M)$ norm of $L^{\alpha/4}f$ with the $L^2(\mathbb{R}^n,M)$ norm of a version of $S_\alpha f$. Finally, the converse inequalities to (3.22) and (3.23) hold in $\mathbb{R}^n$ and also on a unimodular Lie group ([CRTN01]), and we did not consider the corresponding inequalities in the present paper.

Source PDF p. 14: proof line and exact $L^p$ question.

## 2 Counterexample to the printed annular estimate

**Theorem 1** (failure of Lemma 3.4(B)). *Assertion B of the source's Lemma 3.4 is false, even when $n = 1$, $t = 1$, and*

$$d\mu_M(x) = M(x)\,dx = (2\pi)^{-1/2}e^{-x^2/2}\,dx.$$

*In fact, for the standard unit tiling and a sequence of smooth compactly supported functions, the left-hand side stays bounded away from zero while the asserted right-hand side tends to zero.*

*Proof.* Use the tiling centers $x_j^1 = j$. For $k \geq 3$, set $R = 2^k$ and choose the center $x_j^1 = R$. Then

$$Q(R, 2^{k+1}) = [0, 2R], \qquad C_k^{j,1} = [0, R/2] \cup [3R/2, 2R],$$

whereas the mean cube is $Q(R, 2) = [R - 1, R + 1]$.

Choose $f_k \in C_c^\infty(\mathbb{R})$, $0 \leq f_k \leq 1$, so that

$$f_k = 0 \quad \text{on } [R-1, R+1], \qquad f_k = 1 \quad \text{on } [0, 2R] \setminus [R-2, R+2].$$

The cutoff to compact support can be placed outside $[0, 2R]$. Its subtracted mean is exactly zero, and $f_k = 1$ on $C_k^{j,1}$. Consequently

$$\|g_k^{j,1}\|^2_{L^2(C_k^{j,1},\mu_M)} \geq \mu_M([0, R/2]) \longrightarrow \frac{1}{2}. \tag{1}$$

Let $B_R = [R - 2, R + 2]$ and write $Q_R = [0, 2R]$. If $x \in Q_R \setminus B_R$, then $f_k(x) = 1$ and a difference can occur only for $y \in B_R$; for $x \in B_R$, use the crude inner bound $|Q_R| = 2R$. Thus

$$\begin{aligned} N_k &:= \int_{Q_R}\int_{Q_R} |f_k(x) - f_k(y)|^2\,dy\,d\mu_M(x) \\ &\leq |B_R|\mu_M(Q_R \setminus B_R) + 2R\mu_M(B_R) \leq 4 + 2R\mu_M(B_R). \end{aligned} \tag{2}$$

The right-hand side asserted in B, without its fixed constant, is therefore

$$\frac{N_k}{2^k} \leq \frac{4}{R} + 2\mu_M([R-2, R+2]) \longrightarrow 0. \tag{3}$$

Equations (1) and (3) contradict the existence of a uniform constant $\bar{C}$.

Finally, the Gaussian is fully admissible in the source: for $V(x) = x^2/2 + \frac{1}{2}\log(2\pi)$, $(1-\varepsilon)|V'|^2/2 - V'' \to \infty$. Thus the failure is not caused by leaving the paper's measure class. $\square$

## 3 The correct annular estimate

The problem is exactly the scale of the mean. It is taken on the small cube, so no factor $2^{-kn}$ can be gained by "arguing similarly."

**Proposition 1** (corrected B). *With the source's notation, for every $k \geq 1$,*

$$\|g_k^{j,t}\|^2_{L^2(C_k^{j,t},\mu_M)} \leq \frac{C_n}{t^{n/2}} \int_{Q(x_j^t, 2^{k+1}\sqrt{t})}\int_{Q(x_j^t, 2^{k+1}\sqrt{t})} |f(x) - f(y)|^2\,dy\,d\mu_M(x). \tag{4}$$

*The same argument gives, for $1 \leq p < \infty$,*

$$\|g_k^{j,t}\|^p_{L^p(C_k^{j,t},\mu_M)} \leq \frac{C_n}{t^{n/2}} \int_{Q(x_j^t, 2^{k+1}\sqrt{t})}\int_{Q(x_j^t, 2^{k+1}\sqrt{t})} |f(x) - f(y)|^p\,dy\,d\mu_M(x). \tag{5}$$

*Proof.* For $x \in C_k^{j,t}$, Jensen's inequality on the actual mean cube $Q_0 = Q(x_j^t, 2\sqrt{t})$ gives

$$|g^{j,t}(x)|^p \le \frac{1}{|Q_0|}\int_{Q_0} |f(x)-f(y)|^p\,dy = \frac{C_n}{t^{n/2}}\int_{Q_0} |f(x)-f(y)|^p\,dy.$$

Integrate in $x$ over $C_k^{j,t}$ and enlarge both domains to the large cube. This proves (5); $p=2$ gives (4). $\square$

There is also a literal square-difference local $L^p$ form, valid for every $1 \le p < \infty$:

$$\|g_k^{j,t}\|_p^p \le C_n t^{-np/4} \int_{C_k^{j,t}} \left( \int_{Q_0} |f(x)-f(y)|^2\,dy \right)^{p/2} d\mu_M(x).$$

Thus the elementary local averaging step itself is not the difficulty in the intended global $L^p$ problem.

## 4 Repair of the source's main $L^2$ proof

**Theorem 2** (the polynomial loss is harmless)**.** *Replacing Assertion B by Proposition 1 leaves the conclusion of the source's Lemma 3.3, and hence its main $L^2$ theorem, valid.*

*Proof.* Insert (4) into the source's $t$-integral. For fixed $t, x, y$, the number of enlarged cubes of side $2^{k+1}\sqrt{t}$ containing both points is at most $C_n 2^{kn}$. Such a cube can exist only when $|x-y| \le C_n 2^k \sqrt{t}$. Therefore the $k$th annular contribution is at most

$$C2^{kn} \iint |f(x)-f(y)|^2\,d\mu_M(x)\,dy \int_{|x-y|^2/(C_n 4^k)}^{A} t^{-1-(n+\alpha)/2}\,dt,$$

and hence at most

$$C2^{k(2n+\alpha)} \iint_{|x-y|\le C_n 2^k\sqrt{A}} \frac{|f(x)-f(y)|^2}{|x-y|^{n+\alpha}}\,d\mu_M(x)\,dy. \tag{6}$$

This deliberately safe exponent also absorbs the normalization and covering factor discrepancies visible in the printed equations (3.20)–(3.21).

The source's resolvent estimate already multiplies the $k$th contribution by $e^{-c2^k}$. Put $\beta = 2n+\alpha$. The elementary tail bound

$$\sum_{\substack{k\ge 1\\ r\le C_n 2^k\sqrt{A}}} 2^{\beta k} e^{-c2^k} \le Ce^{-c'r}, \qquad r \ge 0,$$

follows by comparing the sum with its first admissible dyadic scale (enlarging $C$ for bounded $r$). Summing (6) therefore gives exactly the exponentially damped fractional difference integral asserted in Lemma 3.3. No later part of the paper needs the false factor $2^{-kn}$. $\square$

## 5 What can and cannot yet be upgraded to $L^p$

One useful part of the $L^p$ extension is automatic. Let $R_t = (I+tL)^{-1}$. It is a positive contraction on $L^1(\mu_M)$ and $L^\infty(\mu_M)$, while the source proves, for disjoint closed $E, F$ at distance $d$,

$$\|\mathbf{1}_F R_t \mathbf{1}_E\|_{2\to 2} \le 8e^{-C_1 d/\sqrt{t}}.$$

Riesz–Thorin interpolation yields the following.

**Lemma 1** (scalar $L^p$ off-diagonal decay). *For $1 < p < \infty$, $p' = p/(p-1)$, and $\theta_p = 2\min\{1/p, 1/p'\}$,*

$$\|\mathbf{1}_F R_t \mathbf{1}_E\|_{p\to p} \le 8^{\theta_p} e^{-\theta_p C_1 d/\sqrt{t}}.$$

*The same bound holds for $tL(I+tL)^{-1}$ between disjoint $E$ and $F$, because its identity term vanishes there.*

What remains is not scalar off-diagonal decay. The intended conclusion is a vector-valued estimate of the form

$$\left\| \left( \int_0^A t^{-1-\alpha/2} |tL(I+tL)^{-1} f|^2 \, dt \right)^{1/2} \right\|_{L^p(\mu_M)} \lesssim \|D_{\alpha,c} f\|_{L^p(\mu_M)}.$$

For $p = 2$, the source squares local estimates, sums disjoint cubes, and uses Fubini. For $p \neq 2$, scalar interpolation does not justify the corresponding $L^p(L^2(dt/t))$ localized estimate. Equivalently, one needs an appropriate vector-valued or $R$-bounded off-diagonal family. The embeddings between $L^p(\ell^2)$ and $\ell^2(L^p)$ point in opposite directions on the two sides of $p = 2$, so neither Minkowski shortcut covers all $p$.

This packet therefore gives a full negative answer to the *literal* printed annular assertion, a complete repair of the $L^2$ theorem, and two genuine $L^p$ ingredients, but it does not claim to settle the intended global square-function comparison.

## 6 Novelty, scope, and human review

A bounded run-index and web/arXiv search on 17 August 2026 used arXiv id 0911.4563, the title and authors, "Lemma 3.4", the displayed annular normalization, and erratum/correction terms. It found copies of the same statement and later adjacent nonlocal $L^p$ Poincaré inequalities, but no erratum, counterexample, or matching repair. Novelty confidence is moderate to high, not conclusive.

Human review should focus on three checks: the drifting-center Gaussian construction in Theorem 1, the corrected mean-cube normalization in Proposition 1, and the safe exponent $2n + \alpha$ in Theorem 2. The distinct global $L^p$ question is explicitly left unresolved.

## References


[1] C. Mouhot, E. Russ, and Y. Sire, *Fractional Poincaré inequalities for general measures*, J. Math. Pures Appl. 95 (2011), 72–84; arXiv:0911.4563.

[2] J. Wang, *A simple approach to functional inequalities for non-local Dirichlet forms*, ESAIM Probab. Stat. 18 (2014), 503–513.